

Validation of the First Law of Thermodynamics in a Pressurized Tank

CHE260 Lab Report by Rooney Cheung, Jady Zeng, Akhilesh Menon, Session PRA0101.

Abstract

Accurate evaluation of energy transfer is essential for validating the First Law of Thermodynamics in real engineering systems where heat loss and non-ideal behaviour occur, such as designing buildings [3] or central heating systems [4]. This experiment applied the First Law ($Q + W = \Delta E$) in a closed, pressurized tank to quantify energy interactions and determine the constant-volume specific heat capacity (c_v). Four trials were performed under varying pressures (40, 70, 80 psig) and target temperatures (40°C , 60°C) for three analyses: determining the air's initial mass, quantify heat loss through the tank walls and plates, and calculating the work done by an internal propeller. These results yielded experimental c_v values ranging from 66.9 ± 3.0 to $123.1 \pm 7.3 \text{ KJ/Kg} \cdot \text{K}$, deviating from the theoretical value of $0.718 \text{ KJ/Kg} \cdot \text{K}$ due to unaccounted heat losses and non-uniform temperature distribution. Moreover, analysis of the energy balance confirmed that the mechanical work done by the propeller was negligible (<1% of total energy input), highlighting the challenges of applying the First Law in practical systems.

Introduction

Thermodynamics is the study of energy and its transformations, governed by the **First Law of Thermodynamics**, which states that energy cannot be created or destroyed, only transferred or changed from one form to another ($Q + W = \Delta E$) [2, 8]. Understanding the conservation of energy is critical for the design and analysis of thermal systems, including engines [5], power plants [6], and chemical reactors [7]. This foundational principle is especially relevant when

analyzing closed, non-flow systems where heat transfer (Q) and work (W) interact to change the system's internal energy (ΔE).

The objective of this experiment was to experimentally validate and apply the First Law to a closed, pressurized air tank system under various conditions of pressure and temperature. First, the **mass of air** within the tank was determined using the Ideal Gas Law under varying pressure conditions, used for subsequent calculations. Second, the system's energy balance was analyzed during a heating and temperature-holding phase to calculate the **rate of heat loss** (Q_{loss}) through the tank walls and determine the **specific heat capacity of air at constant volume** (c_v). Third, mechanical energy input was explored by quantifying the **work done by a propeller** and evaluating its significance relative to the total heat added to the system. By measuring and accounting for these key energy components—internal energy change, heat transfer, and work—this experiment provides a comprehensive demonstration of the conservation of energy principle in a controlled thermal system [1, 2].

Experimental Method

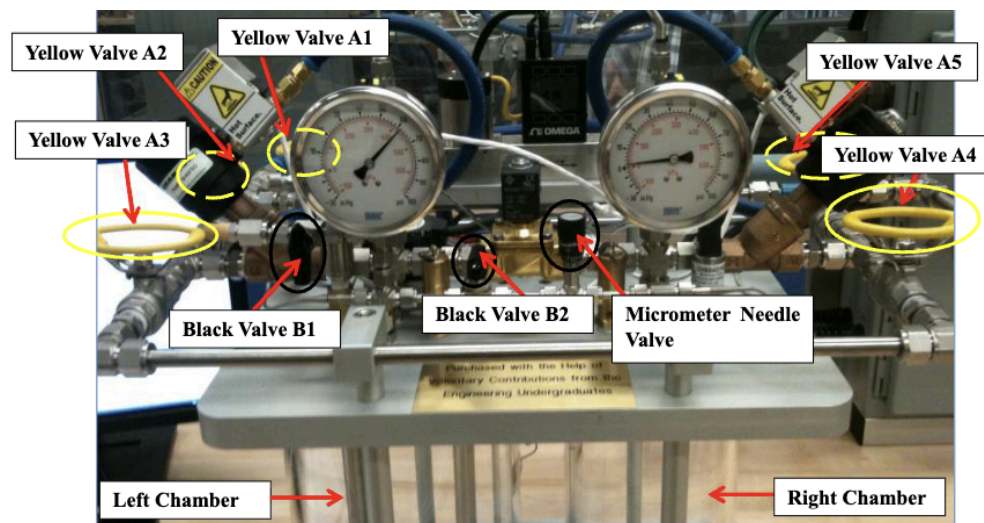


Figure1: Experimental setup with labeled valves [2]

Part 1: Determining the Mass in the Left Tank: Air in the tank is pressurized to analyze the change in state. First, record the ambient pressure (P_{atm}) using the lab's manometer. Click the **'Start Collecting Data'** button on the LabVIEW program. Open Valve **A2**, and using LabVIEW open the **Left Solenoid valve** and set the mass flowrate to 50g/min. When the pressure reaches 40 psig, close the Left Solenoid Valve, the A2 Valve, and set the mass flowrate to 0g/min. Wait until the pressure (P) and temperature (T) stabilize (i.e., until values on the graphs remain constant), then stop data collection and save as **Lab 2 - Part 1a**. Note: Do not empty the tank before proceeding to Part 2. *Repeat Part 1 procedure for the remaining three pressure and temperature conditions.*

Part 2: Determining the Heat Loss and Specific Heat Capacity: Air is heated to a target temperature and maintains it to analyze energy input and loss. Using the LabVIEW program, click the **'Start Collecting Data'** button, set the **Target Temperature** to 40°C , and click the **'Hold'** button to power the heater. Monitor the system until the left tank temperature reaches the 40°C target, then allow the system to run for an additional **5 minutes** to ensure a steady-state condition for heat loss analysis. Stop data collection and save data as **Lab 2 - Part 2a**. To cool-down, turn off the heaters by clicking Hold, and reset data collection to monitor the tank temperature. Open the Left Solenoid valve and the **"Bar"** valve behind the left tank. Once the tank had cooled to 30°C (40°C for all other trials) and the temperature stabilizes below this point, close the Left Solenoid valve and the "Bar" valve. Evacuate the left tank by opening and closing **Valve B1**. *Repeat Part 2 procedure for the remaining three pressure and temperature conditions.*

Part 3: Determining the Work Done by the Propeller

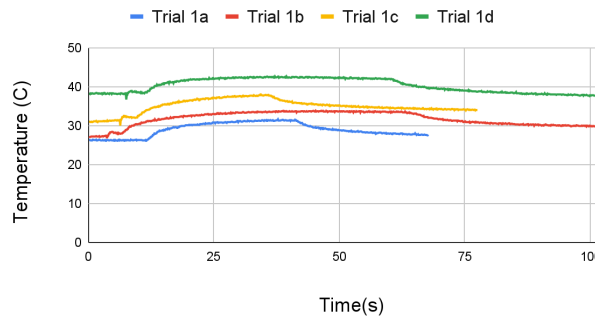
The work done by the propeller was assessed to evaluate its energy contribution to the system.

Propellers run continuously during the experiment, no procedures are required.

Results and Discussion

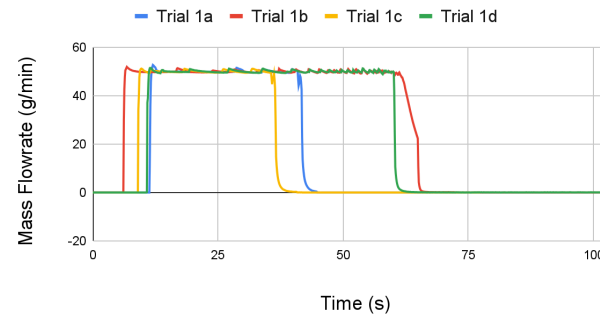
[1] Determining the Mass in the Left Tank: The experiment produced measurements for mass flow rate, pressure, and temperature over time for the left tank.

Pressurizing Gas Temperature VS. Time



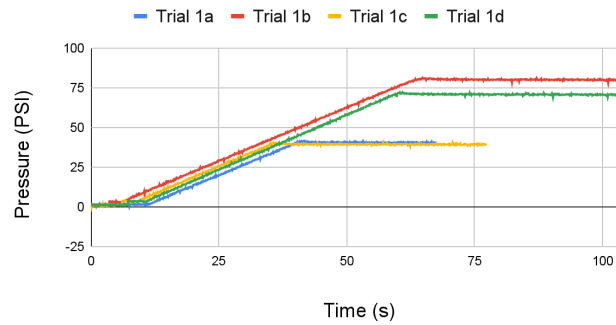
(a)

Mass Flowrate VS. Time



(b)

Gas Pressure VS. Time



(c)

Figure 1a.: Temperature as a function of time while the tank is being pressurized. Figure 1b.: Mass flow rate as a function of time, set to constant 50g/min, exhibited as a square pulse before closing the solenoid, returning flow rate to 0g/min. Figure 1c.: Gas pressure as a function of time to pressurize the tank to constant 40psi (trial 1a, 1c), 70psi (trial 1d) and 80psi (trial 1b).

$$m_{\text{left tank}} = m_{\text{added to tank}} \left[1 + \frac{1}{\left(\frac{P_2 T_1}{P_1 T_2} \right) - 1} \right] - [1] \text{ where Absolute Pressure, } P_{1,2} = P_{\text{Atm}} + P_G ;$$

The mass added to the tank can be taken by the integral of the mass flow rate function of time within the time interval of which the valve opens and closes. In Figure 1b, the mass flow rate dm exhibits a square pulse, peaking suddenly and remaining approximately constant throughout this interval, with small variations (± 2 g/min) likely due to experimental uncertainty. Thus, the square pulse shape's area, representing the net mass of the system, is determined by:

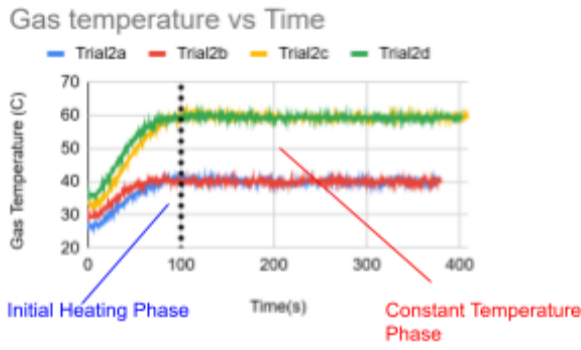
$$\text{Mass of air added, } m_{\text{added to tank}} = dm \cdot t - [2]$$

P_1 , P_2 , T_1 , T_2 from Equation 1, were all derived from the pressure and temperature plots as a function of time, taken where they become constant as exemplified in Figure 1a (specified values in Appendix Table B). The uncertainties of these values are estimated by the fluctuations in the plot, taken as the magnitude of the maximum deviated value from the average across the time period. As a result, the total mass of air in the left tank for each trial is:

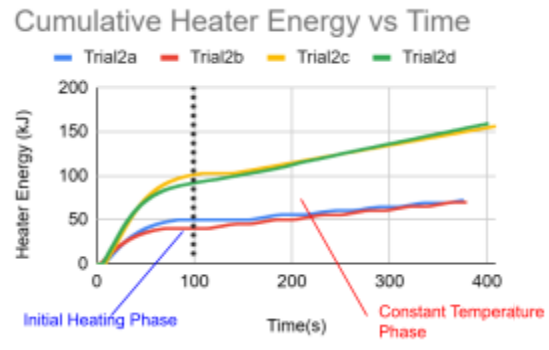
Trial	Mass (kg)
40 psig & 40 °C	0.04 ± 0.01
80 psig & 40 °C	0.06 ± 0.02
40 psig & 60 °C	0.03 ± 0.01
70 psig & 60 °C	0.05 ± 0.01

Table 1: All trials' total mass of air in the left tank

[2] Determining the Heat Loss in the Left Tank and Specific Heat Capacity: The experiment measured temperature, total heat added over the time period, and the outer temperature after initial heating. Uncertainty values are determined by fluctuations in graphical data.



(a)



(b)

Figure 2a.: Gas Temperature vs time, it separates into the initial heating phase, indicated by the left of dotted lines, blue label ($\sim < 100s$) with constant increase, and a constant temperature phase, indicated by the right of dotted lines, red label ($\sim > 100s$). Figure 2b.: Cumulative Heater Energy vs Time, initial heating phase with non linear energy input and constant temperature phase is equal to Figure 2a. For trials 2a, 2b, the heater toggles ON/OFF, for trials 2c, 2d, the heater is on constantly for a linear slope.

Note that the time interval for each trial's 2-phase is slightly different. Assuming the heat loss during heating is negligible, then the heat used to increase the gas temperature, Q_{heat} is equivalent to the heat in the Initial Heating Phase of Figure 2b. By conservation of energy,

$$Q_{\text{loss}} = Q_{\text{total}} - Q_{\text{heat}} - [3]$$

Where Q_{loss} is the heat loss from the tank, Q_{total} is the total heat input throughout the process. (See Appendix)

Trial	Time Interval, $[t_{\text{start}}, t_{\text{end}}]$ (s)	Actual Gas Temperature, T_{gas} ($^{\circ}\text{C}$)	Heat Loss, Q_{loss} (kJ)
40 psig & 40 $^{\circ}\text{C}$	$[84.5, 375.9] \pm 0.5$	40.0 ± 0.2	22.3 ± 0.1
80 psig & 40 $^{\circ}\text{C}$	$[70, 379.3] \pm 0.5$	40.0 ± 0.2	29.8 ± 0.1
40 psig & 60 $^{\circ}\text{C}$	$[109.0, 409.0] \pm 0.5$	59.5 ± 0.2	53.4 ± 0.1
70 psig & 60 $^{\circ}\text{C}$	$[103.4, 403.6] \pm 0.5$	59.5 ± 0.2	66.7 ± 0.1

Table 2: All Trials' starting time, ending time, actual gas temperature and heat loss associated.

Heat loss, Q_{loss} can be further divided into heat loss through the cylindrical wall, Q_{wall} , and heat loss through the top and bottom plates, Q_{plates} , this can be represented as:

$$Q_{loss} = Q_{plates} + Q_{wall}, Q_{plates} = Q_{loss} - Q_{wall} \quad [5]$$

And Fourier's Law of Heat Conduction for the rate of Q_{wall} is given by [1]:

$$Q_{wall}' = 2k\pi l \frac{\Delta T}{\ln(\frac{r_2}{r_1})} = A\Delta T, A \approx 3.65 \pm 0.07 \frac{J}{sK}, \quad [6]$$

Where A [J/sK] is a constant dependent on tank geometry and material (See Appendix). ΔT [K]

is the temperature drop across the wall. Due to rigorous mixing, the inner wall temperature is

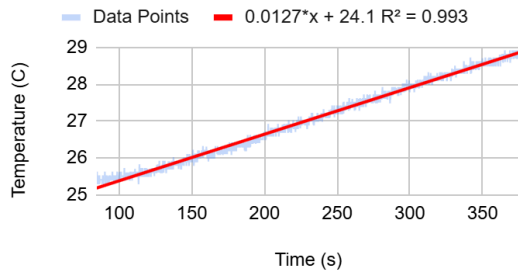
assumed to be equal to the gas temperature, i.e. $T_{in} \approx T_{gas}$. The outer wall temperature, T_{out}

demonstrates a linear trend throughout each trial and is fitted against the linear line $y = mx + b$,

Trial	Linear Temperature Function (K)	R^2	Residual Deviation, σT_{out}
40 psig & 40 °C	$T_{out}(t) = 0.0127t + 24.1 + 273.15$	0.993	± 0.1
80 psig & 40 °C	$T_{out}(t) = 0.0101t + 27.1 + 273.15$	0.989	± 0.1
40 psig & 60 °C	$T_{out}(t) = 0.0214t + 28.5 + 273.15$	0.994	± 0.2
70 psig & 60 °C	$T_{out}(t) = 0.0182t + 32.3 + 273.15$	0.994	± 0.1

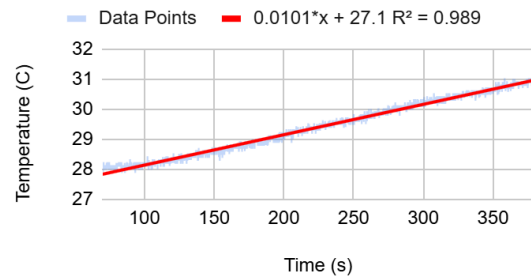
Table 3: All Trials' Outer Wall temperature Linear fits, R^2 value, and the residual deviation

Part 2a Outer Temperature vs. Time



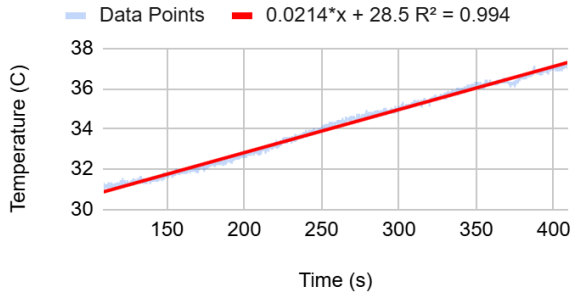
(a)

Part 2b Outer Temperature vs Time



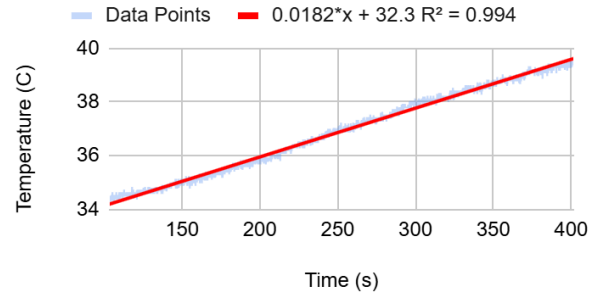
(b)

Part 2c Outer Temperature vs. Time



(c)

Part 2d Outer Temperature vs. Time



(d)

Figure 3abcd, Linear Regressions for each part's outer temperature, note the red line matches the data trend appropriately, with 3a: $R^2 = 0.993$, $m = 0.0127$ °C/s, $b = 24.1$ °C; 3b: $R^2 = 0.989$, $m = 0.0101$ °C/s, $b = 27.1$ °C; 3c: $R^2 = 0.994$, $m = 0.0214$ °C/s, $b = 28.5$ °C; 3d: $R^2 = 0.994$, $m = 0.0182$ °C/s, $b = 32.3$ °C.

Overall, the R^2 values are ≈ 1 , indicating a suitable fit. As a result, ΔT can be expressed as:

$$\Delta T(t) = T_{outer}(t) - T_{in} - [7]$$

To find total heat loss, Equation 7 is integrated over the time interval, thus for each case,

$$Q_{wall} = \int_{t_{start}}^{t_{end}} 2k\pi l \frac{T_{outer}(t) - T_{in}}{\ln(\frac{r_2}{r_1})}, \text{ note } T [K] = T[C] + 273.15 - [8]$$

Trial	Q_{loss} (kJ)	Q_{wall} (kJ)	Q_{plates} (kJ)
40 psig & 40 °C	22.3 ± 0.1	13.7 ± 1.2	8.6 ± 1.2
80 psig & 40 °C	29.8 ± 0.1	12.0 ± 1.0	17.8 ± 1.2
40 psig & 60 °C	53.4 ± 0.1	27.8 ± 2.3	25.6 ± 2.3
70 psig & 60 °C	66.7 ± 0.1	24.7 ± 2.0	42.0 ± 2.0

Table 4: All trials' total heat loss, heat loss through the wall, and through the plates

It is important to note that these values contain possible errors, due to the following reasons: 1.

Ideal Gas Law assumes steady state conditions, however, it is notable that temperature is still non-uniformly distributed due to the constant ON/OFF cycle of the heater, as a result, the actual

temperature after heating is not accurately determined. 2. The assumption of no heat loss during heating impacts the actual heat loss contributes to the heat loss throughout the process.

To determine the specific heat capacity of air, the heat in the initial heating phase in Figure 2b, Q_{heat} , initial mass, m found in Part 1 and the temperature change, ΔT is considered. Due to a rigid tank, neglecting work done, the First Law can be rearranged to give:

$$c_v = \frac{Q}{m\Delta T} \quad [9]$$

Trial	Specific Heat Capacity, c_v (kJ/kg K)
40 psig & 40 °C	106.5 ± 6.0
80 psig & 40 °C	66.9 ± 3.0
40 psig & 60 °C	123.1 ± 7.3
70 psig & 60 °C	79.0 ± 3.3

Table 5: All trials' resultant specific heat capacity, with unusual high values.

Note that these values significantly differ from the theoretical value of $c_v = 0.718$ (kJ/kg K) [2], which is attributed to other undetermined heat loss mechanisms or sensor inaccuracies.

[3] Work Done by the Propeller and Rate of Rise in Temperature

Work done by the propeller is equivalent to its generated power: $W = P_2 \Delta t$ – [10]

where P_2 (fan power) can be determined by the equation $P_2 = P_1 \frac{\rho_2}{\rho_1} \left(\frac{n_2}{n_1}\right)^2$ – [11]

and gas density $\rho_2 = \frac{p_2}{R_{\text{air}} T_2}$ – [12] from the Ideal Gas Law.

Since $\frac{dW}{dt} = P_2$, the rate of temperature rise due to work is $\frac{dT}{dt} = \frac{P_2}{mc_v}$ – [13] under the

assumption that the system is adiabatic ($W = \Delta U$, $W = mc_v \Delta T$).

The mass and c_v values for each trial is reported in *Table1* and *Table5* respectively.

Trial	Density, ρ_2 (kg/m ³)	Power, P_2 (W)	Work (J)	$\frac{dT}{dt}$ (K/s)
40 psig & 40 °C	4.37 ± 0.02	0.572 ± 0.002	166.6 ± 0.7	$(1.5 \pm 0.1) \times 10^{-4}$
80 psig & 40 °C	7.51 ± 0.02	0.982 ± 0.002	303.7 ± 0.8	$(2.6 \pm 0.2) \times 10^{-4}$
40 psig & 60 °C	4.18 ± 0.01	0.546 ± 0.002	163.9 ± 0.6	$(1.4 \pm 0.1) \times 10^{-4}$
70 psig & 60 °C	6.56 ± 0.01	0.858 ± 0.002	257.7 ± 0.7	$(2.2 \pm 0.1) \times 10^{-4}$

Table 6: All trials' gas density, fan power, work by propeller, and rate of temperature increase

Calculating the percentage ratio $\frac{W_{propeller}}{Q_{heat}}$ – [14] using the values in *Table 6* and Q_{heat} , the work

done by the propeller is below 1% of the heat added, demonstrating negligible influence and validating that propeller work can be ignored in earlier calculations in the report.

Conclusion

The First Law of Thermodynamics was applied to analyze the energy balance of a closed, pressurized air tank under varying operational conditions, allowing for quantification of three key energy terms: internal energy change (ΔU), heat transfer (Q), and shaft work (W_{shaft}). The heat loss analysis demonstrated that for the 70 psig and 60 °C trial, the total heat loss was the greatest (66.7 ± 0.1 KJ), indicating a strong dependence of heat loss on the temperature difference between the gas and the surroundings. Furthermore, the calculated ratio of work done by the propeller to the heat added was consistently below 1% (ranging from 0.16% to 0.76%), decisively confirming that the mechanical work was a negligible energy component compared to the heater energy input. However, the experimentally determined specific heat capacity values (c_p) were significantly higher than the theoretical value of 0.718 KJ/Kg • K. This large deviation is attributed to systematic errors, primarily the assumption of uniform internal

temperature, which was likely violated due to the cyclic nature of the PID heater control, and unaccounted heat losses through components other than the main cylindrical wall and plates, such as the mounting hardware and wiring. Future studies should employ a more advanced calorimetry approach or integrate more temperature sensors to better characterize the internal thermal gradients and improve the accuracy of the c_v calculation.

Citations

- [1] *Lab 2 – First Law of Thermodynamics, CHE 260: Thermodynamics and Heat Transfer*, University of Toronto, Toronto, ON, Canada, 2025.
- [2] S. Chandra, *Energy, Entropy and Engines: An Introduction to Thermodynamics*, 1st ed. Hoboken, NJ: John Wiley & Sons, 2016.
- [3] I. Uriarte, A. Erkoreka, P. Eguia, K. Martin-Escudero, “Estimation of the Heat Loss Coefficient of Two Occupied Residential Buildings through an Average Method,” *Energies*, vol. 13, no. 21, Art. 5724, Nov. 2020, . [Accessed: 26-Oct-2025]
- [4] I. Giurca, “Calculation Of Heat Loss Through The Pipes Of The Interior Central Heating System,” *Journal of Applied Engineering Sciences*, vol. 5, no. 2, pp. 29-36, Dec. 2015, doi:10.1515/jaes-2015-0018. [Accessed: 26-Oct-2025]
- [5] J. Demuyne, “Heat Loss Comparison Between Hydrogen, Methane,...,” *Procedia Engineering*, vol. 42, 2012, doi:10.1016/S1876-6102(12)01438-5. [Accessed: 26-Oct-2025]
- [6] V. K. Dwivedi, S. K. Gupta, H. Verma, N. Yadav, H. Yadav, “Improvement in Energy Efficiency & Heat Loss Minimization during Boiler Operation: A Case Study,” in *IOP Conf. Ser.: Mater. Sci. Eng.* vol. 1116, no. 1, p. 012044, May 2021, doi:10.1088/1757-899X/1116/1/012044. [Accessed: 26-Oct-2025]
- [7] P. Harriott, “Heat and Mass Transfer in Reactors,” in *Chemical Reactor Design*, 1st ed., CRC Press, 2002, ch. 7, pp. 50. [Accessed: 26-Oct-2025]
- [8] ScienceDirect, "First Law of Thermodynamics," *ScienceDirect Topics*. [Online]. Available: <https://www.sciencedirect.com/topics/chemistry/first-law-of-thermodynamics>. [Accessed: 26-Oct-2025]

Appendix

Constant	Value
Manufacturer Fan Power (P_1)	0.001 hp = 0.7457 W
Air density (ρ_1)	1.293 kg/m ³
Manufacturer Pump Shaft Speed (n_1)	4200 rpm
Operating Pump Shaft Speed (n_2)	2000 rpm
Air Gas Constant (R_{air})	0.287 kJ/kg K
Atmospheric Pressure (P_{atm})	29.9 $\pm$ 0.1 inHg = 101.25 $\pm$ 0.05kPa

Table A: Experimental constants

Trial	Average mass flowrate (g/min)	Time interval (s)	Initial Pressure, P_1 (psi)	Final Pressure, P_2 (psi)	Initial Temperature, T_1 (C°)	Final Temperature, T_2 (C°)
40 psig & 40 °C	50 $\pm$ 2	30.2 $\pm$ 0.1	2 $\pm$ 1	40 $\pm$ 1	26.3 $\pm$ 0.2	27.8 $\pm$ 0.3
80 psig & 40 °C	50 $\pm$ 2	57.0 $\pm$ 0.1	0.7 $\pm$ 0.6	80 $\pm$ 2	27.2 $\pm$ 0.2	29.9 $\pm$ 0.2
40 psig & 60 °C	50 $\pm$ 2	27.3 $\pm$ 0.1	1 $\pm$ 2	39 $\pm$ 1	31 $\pm$ 1	34.1 $\pm$ 0.2
70 psig & 60 °C	50 $\pm$ 2	49.3 $\pm$ 0.1	1.2 $\pm$ 0.6	71 $\pm$ 1	38 $\pm$ 2	36.8 $\pm$ 0.3

Table B: All trials' quantitative values when tank is pressurized

Trial	Q _{total} (kJ)	Q _{heat} (kJ)
40 psig & 40 °C	72.2 ± 0.1	49.9 ± 0.1
80 psig & 40 °C	69.8 ± 0.1	40.0 ± 0.1
40 psig & 60 °C	156.1 ± 0.1	102.7 ± 0.1
70 psig & 60 °C	159.2 ± 0.1	92.5 ± 0.1

Table C: All trials' cumulative total heat value and heat used to heat up the gas.

A, Tank material, geometric constant calculation [1]:

- k is the conductive heat transfer coefficient [w/mK], since the walls are acrylic, $k \approx 0.185 \pm 0.015$ [w/mK]. The length of the cylinder, $l = 285.75 \pm 0.03$ mm.
- The outer diameter of the cylinder is $d_2 = 8'' \pm 0.045''$ with a wall thickness of $w_{wall} = 3/8'' \pm 15\%$, the following relations can be deduced:

$$r_2 = \frac{d_2}{2} = 0.1016 \pm 0.001 [m]$$

$$r_1 = r_2 - w_{wall} = 0.09275 \pm 2 \cdot 10^{-3} [m]$$

- $r_2 = 0.1016 \pm 0.001$ m is the outer radius of the container, $r_1 = 0.09275 \pm 2(10^{-3})$ m is the inner radius of the container.

Uncertainties for each Equations

[1]

$$\delta m_{left\ tank} = m_{left\ tank} \sqrt{\left(\frac{\delta m_{added}}{m_{added}}\right)^2 + \frac{1}{\left(\frac{P_2 T_1}{P_1 T_2} - 1\right)^2} \left[\left(\frac{\delta P_2}{P_2}\right)^2 + \left(\frac{\delta T_1}{T_1}\right)^2 + \left(\frac{\delta P_1}{P_1}\right)^2 + \left(\frac{\delta T_2}{T_2}\right)^2 \right]}$$

$$[3] \delta m = m \sqrt{\left(\frac{\delta dm}{dm}\right)^2 + \left(\frac{\delta \Delta t}{\Delta t}\right)^2}$$

$$[4] \delta Q_{loss} = \sqrt{\delta Q_{total}^2 + \delta Q_{heat}^2}$$

$$[6] \delta A = A \sqrt{\left(\frac{\delta k}{k}\right)^2 + \left(\frac{\delta l}{l}\right)^2 + \left(\frac{\left(\frac{\delta r_2}{r_2}\right)^2 + \left(\frac{\delta r_1}{r_1}\right)^2}{\left(\ln\left(\frac{r_2}{r_1}\right)\right)^2}\right)}$$

$$[8] \delta Q_{wall} = Q_{wall} \sqrt{\left(\frac{\delta t_{start}^2 + \delta t_{end}^2}{(t_{end} - t_{start})^2}\right) + \left(\frac{\sigma T_{outer}^2 + \delta T_{in}^2}{(T_{outer, avg} - T_{in})^2}\right) + \left(\frac{\delta A}{A}\right)^2}$$

$$[9] \delta c_v = c_v \sqrt{\left(\frac{\delta Q}{Q}\right)^2 + \left(\frac{\delta m}{m}\right)^2 + \left(\frac{\delta \Delta T}{\Delta T}\right)^2}$$

$$[10] \delta W = W \sqrt{\left(\frac{\delta P_2}{P_2}\right)^2 + \left(\frac{\delta \Delta t}{\Delta t}\right)^2}$$

$$[11] \delta P_2 = P_2 \left(\frac{\delta \rho_2}{\rho_2}\right)$$

$$[12] \delta \rho_2 = \rho_2 \sqrt{\left(\frac{\delta p_2}{p_2}\right)^2 + \left(\frac{\delta T_2}{T_2}\right)^2}$$

$$[13] \delta \frac{dT}{dt} = \frac{dT}{dt} \sqrt{\left(\frac{\delta P_2}{P_2}\right)^2 + \left(\frac{\delta m}{m}\right)^2 + \left(\frac{\delta c_v}{c_v}\right)^2}$$

$$[14] \delta \frac{W_{propellor}}{Q_{heat}} = \frac{W_{propellor}}{Q_{heat}} \sqrt{\left(\frac{\delta W_{propellor}}{W_{propellor}}\right)^2 + \left(\frac{\delta Q_{heat}}{Q_{heat}}\right)^2}$$

$$\delta r_1 = \sqrt{\delta r_2^2 + \delta w_{wall}^2}$$